Your first useful result

A five-to-ten-minute starting exercise using clearly labeled synthetic data

1 Extract and launch

Open the v0.11.0b7 Research Beta release and download **tpd-spectra-windows-x64.zip**. No GitHub sign-in or Python installation is needed. Extract the whole ZIP, keeping **docs** and **examples** beside **TPD-Spectra-Studio.exe**. Double-click the executable. The noisy two-peak demo opens with its first peak selected. Check VERSION.json for the source commit and the separate ZIP checksum for integrity. The unsigned beta must comply with your institution’s software policy.

This prerelease includes the two-peak improvements merged in PR #37. Earlier development artifacts with the same b7 version label may lack them. Experimental validation and wider user testing remain ongoing.

2 Open the clean example

Click **Browse examples…**. Keep **first order single peak** and open it. You should see 801 points and a named full-trace interval, 50–450 °C. The chooser supplies the documented 1 K/s ramp. These are simulated rates, not measurements.

3 Read an energy estimate

Open **Kinetic Analysis** and select **Redhead**. Check heating rate **1 K/s** and assumed prefactor **1e13 s^{-1}**. The automatic preview should be approximately **1.2 eV**, within 2.5% of the simulation input. This first-order approximation has no measured confidence interval.

4 Keep the result

Click **Save labeled result…** and give it a meaningful name. Use **Report / Export…** to select that run and choose a destination. Open the exported PDF. Use **Save project…** to retain the whole workspace for later.

Need help? The beta includes a feedback form under **Help > Report a bug / Send feedback**. Even a short note about an unclear control or unexpected result is useful. This walkthrough uses synthetic data, so there is no need to share your measurements.

Prepare and compare

Keep the dataset, named peak and processing recipe visible

Bring your own spectrum

Use **Open Spectrum…** for a CSV. Confirm temperature and time units, heating rate and signal meaning from acquisition records. The synthetic CSV format is **Time_ms, Signal, Temperature**, with temperature in °C. Inspect the import report; do not change cleaning thresholds merely to force bad data through.

Define a physically useful interval

In **Spectra & Preparation**, enter a peak name and From/To temperatures, then **Add peak**. Or use **Drag interval** with plot pan/zoom off. Save edits with **Update peak**. The persistent Dataset and Peak selectors control all three stages. Include adequate tails; avoid unmodeled neighbors.

Separate display choices from analysis

Normalization changes the display only. Smoothing initially previews the plot. **Apply smoothing / baseline to analysis** records an analysis recipe; Undo and Reset are available. Changing bounds or processing marks previous results stale, while saved runs retain their original settings.

Explore a hypothesis

In **Modeling & Order Identification**, simulate parameters or select candidate orders and **Compare models**. Start with the first-order candidate on the clean first-order example. Inspect fit residuals, convergence and identifiability warnings, not only the lowest AICc. A good curve match does not identify a unique mechanism.

Fixed zero-order fitting: the new method searches the depletion edge. Include rising signal and at least three samples after exhaustion. Prefactor and scale represent conditional ranges, not unique estimates or confidence intervals. Review residuals; numerical completion does not prove the model.

The **Examples** menu separates simulated spectra with known inputs from cited, publication-derived data with source and limitations. Neither proves a kinetic model. See **examples/README.md**.

Fit and inspect uncertainty

Simulation, model comparison and a final kinetic fit are separate steps

Transfer a candidate deliberately

Open **Parameters & models**, select a converged comparison row and choose **Use selected model in Kinetic Analysis**. This transfers a hypothesis and initial guesses. Check the method, ramp, order and interval, then press **Calculate / Fit selected peak**. Inspect residuals and **Assumptions & provenance** before saving.

Choose a method that fits the question

Leading edge: apparent slope energy on a rising interval with little coverage change. On the clean reference, try 90–110 °C; expect about 1.2 eV. Do not use the entire peak. This does not assign zero order.

Redhead: first-order estimate with an assumed prefactor and linear ramp.

Polanyi–Wigner: single-state numerical fitting. Energy, prefactor and amplitude may be poorly identifiable. The fitter normalizes initial coverage to one; absolute coverage is not measured here.

Try Monte Carlo after comparison

Return to Modeling, complete a current comparison, choose a seed/count, then **Run Monte Carlo**. For a short software check use first order only, multistart off, 20 replicates and seed 42. The visible bar shows count, percentage, elapsed time and an estimated remainder. **Cancel** preserves prior valid results. Completion separates fit failures from incomplete rankings. Conditional ranges require 100 successful refits and are not coverage-validated.

What Monte Carlo means here

This is **parametric refitting** with iid Gaussian noise around the lowest-AICc converged reference and its residual noise estimate. It is not microscopic kinetic Monte Carlo or Bayesian MCMC. Rank frequencies describe conditional stability, not probabilities of mechanisms. Drift, correlated noise, calibration and model uncertainty are not included.

Science before presentation

Synthetic recovery is a software check. The clean benchmarks use known-order, truth-informed starts; they do not demonstrate robust blind inference. Compare real results with an independent calculation and suitable reference measurements. Keep acquisition conditions, model assumptions and limitations with every exported result.

Fit overlapping components

Explicit independent physical populations on one full experimental spectrum

Define physical hypotheses

In Modeling choose **Fit 2 Peaks…**. Verify the rate, click approximate positions (or type °C values), then **Fit Both Peaks**. New rows fix n=1, prefactor 1e13 s^{-1} and coverage one. Redhead energy guesses and ±30% bounds are heuristic search aids, not uncertainty intervals. **Advanced settings** reveals fixed kinetics, bounds, baseline and optimizer controls. **Overlapping components…** supports one to three rows.

Try the three-state synthetic reference

Open **three overlapping peaks**; in Advanced add a third row. Fix first order, prefactor **1e13 s^{-1}**, coverage one; energies **1.185, 1.265, 1.345 eV**; bounds **1.14–1.22, 1.22–1.30, 1.30–1.38 eV**; 1 K/s; multistart off; 300 evaluations. Choose **Fit 3 Components Together**. This known-input software exercise does not establish blind physical identification.

Inspect the whole fitted signal

The **entire dataset** is fitted. Blue dashed and green dash-dot contributions, orange sum, dotted baseline and observation-minus-sum residuals are distinct. **Baseline-only, poor-fit or collapsed-component** titles/status require review even after convergence. Read **Assumptions & warnings**. Count comparison uses prefixes of the entered rows; rankings are heuristic.

Published rates require confirmation

Zaki D_2O/Fe_3O_4 suggests **3 K/s**: explicitly accept/reject it. Opening published data never silently changes the rate. Confirmation applies to Modeling only; verify Kinetic Analysis separately. The −20 °C first-order seed is about 0.67 eV. Correlated residuals still limit this digitized stress test; two states and experimental energies are not validated.

Retain assumptions and uncertainty limits

Save project… retains definitions/results. **Export component fit…** writes parameters, curves, residuals, provenance and PNG/SVG. Local intervals condition on fixed kinetics and iid noise; ambiguity withholds them. Component Monte Carlo and joint zero-order fitting are unavailable. See **docs/MULTICOMPONENT_FITTING.md**.

Save, share and report

Choose what leaves the application and inspect it before sharing

Review results and choose the destination

Use **Save labeled result…** for each result you want to retain. **Report / Export…** shows the saved-run selection; use Ctrl/Shift for multiple rows. Click **Choose destination and generate report…**. A fresh folder receives PDF, parameter/curve CSV, JSON, PNG and SVG. Existing report folders are preserved. The dialog previews selected content, not the rendered PDF.

Save a portable project

Save project… writes a .tpd.json file containing imported arrays, named peaks, settings, recipes, saved results and modeling sessions. **Open project…** restores them. Projects and analysis reports contain data and provenance; they are not privacy-filtered diagnostic reports.

Send useful feedback without a GitHub account

Open **Help > Report a bug / Send feedback**, choose a category, describe the problem and optionally add steps or your email. Click **Update report preview**. Inspect/edit the exact text. **Copy Diagnostic Report** works offline; **Save Diagnostic ZIP** saves only the report and an explanatory README. No spectra or arbitrary files are attached.

After checking the review box, **Send Feedback by Email** opens a draft to **karol.j.cieslik@gmail.com**. Attach the ZIP manually if wanted and decide whether to send. If no mail application opens, copy the report into your preferred mail service. **Report on GitHub** opens a public issue draft; review it before submission.

About and contribution

Dr. Karol Cieślik created and maintains TPD Spectra Studio. About shows the version/build, contact, source, fork and license information. Suggestions, scientific discrepancies and contributions are welcome.
github.com/karol-cieslik/TPD-Spectra-Analysis

No experimental data is required for feedback. Share only material you have permission to share. Private debugging permission does not authorize publication or redistribution. See the optional tester checklist for a short, low-pressure review.